



Extending RotatE: Interpretability and Multi-Relational Modeling

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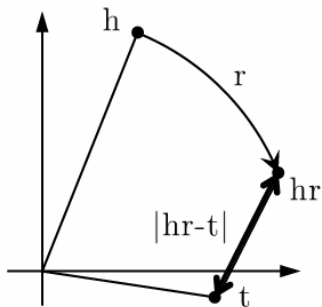
Introduction & Motivation



Motivation & Problem

Knowledge Graphs (KGs) are critical for **reasoning** tasks in NLP, recommendation, and biomedical domains.

RotatE models relations as **rotations** in the complex plane.



RotatE

$$-\|\mathbf{h} \circ \mathbf{r} - \mathbf{t}\|^2$$

$$\mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{C}^k, |r_i| = 1$$

Published as a conference paper at ICLR 2019

ROTATE: KNOWLEDGE GRAPH EMBEDDING BY RELATIONAL ROTATION IN COMPLEX SPACE

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Motivation & Problem

Limitations:

- Struggles with non-commutative relations.
- Poor handling of 1-to-N, N-to-1, N-to-N patterns.
- Biased by relation imbalance.
- Interpretability is essential in high-stakes domains.

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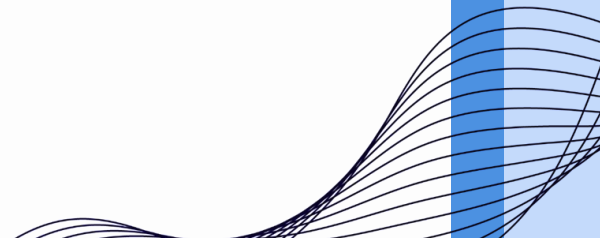
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Research Objectives

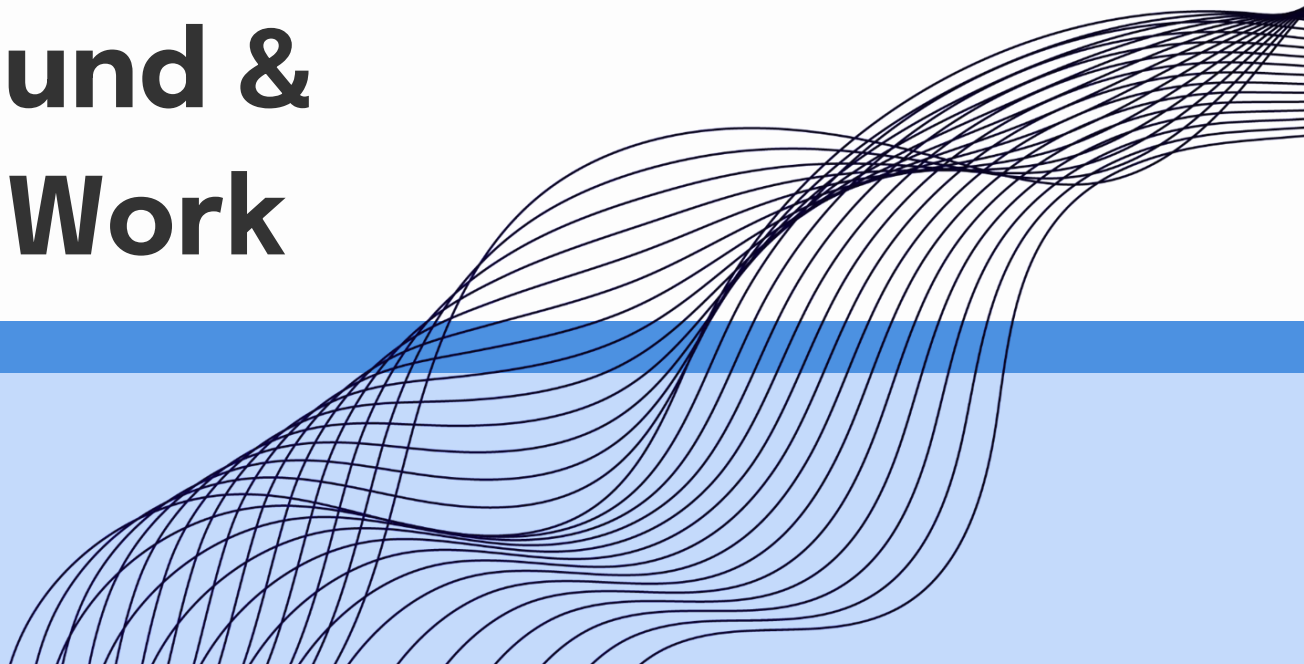
Extend RotatE for:

- Better multi-relational modeling
- Improved training stability
- Interpretability of embeddings
- Evaluate on FB15k/FB15k-237 benchmark
- Compare multiple RotatE variants systematically



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Background & Related Work



Background – RotatE

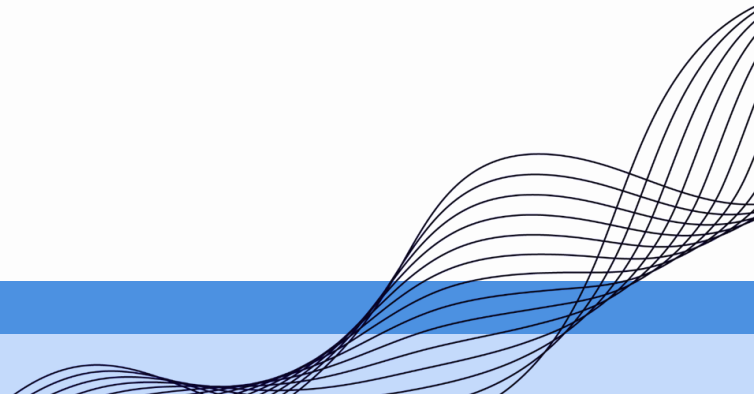
Embeds entities & relations in complex space.

Models:

Model	Score Function	Symmetry	Antisymmetry	Inversion	Composition
RotatE	$-\ h \circ r - t\ $	✓	✓	✓	✓

Strength: Elegant geometric interpretation

Weakness: Limited for complex relation types



Related Extensions

RotatE-CT

adds translation $\rightarrow$ non-commutative relations

MRotatE

entity rotation $\rightarrow$ multi-fold relations.

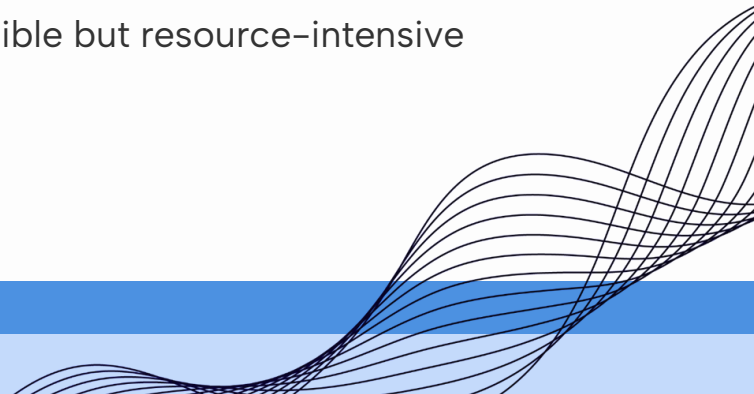
Related Works

- **Early KGE Models**

- **TransE** → Simple translations; struggles with 1-to-N / N-to-N.
- **DistMult, ComplEx** → Bilinear & complex embeddings; capture asymmetry but miss non-commutativity.

- **Beyond 2D Rotations**

- **Rotate3D** → Richer geometric transformations, higher-order patterns; higher computational cost.
- **QuatE / QuatD** → Quaternion embeddings; flexible but resource-intensive



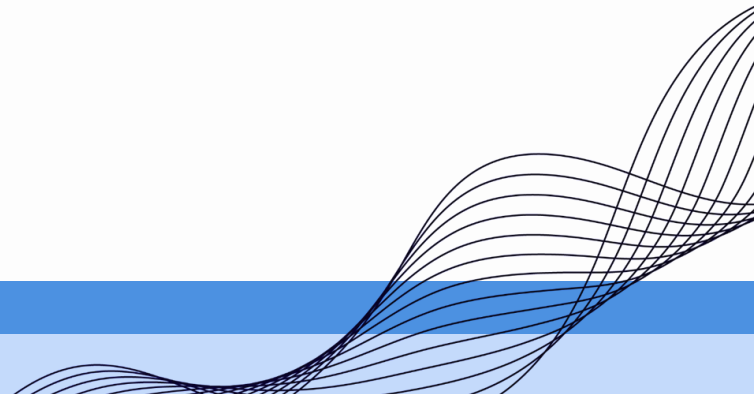
Related Works

- **Alternative Approaches**

- **TuckER** → Tensor-based; very expressive but memory-heavy.
- **R-GCN** → Graph convolutional; strong predictive power, low interpretability.

→ **Gap**: Many models improve accuracy but often trade off **scalability** or **interpretability**.

Our work builds on **rotational embeddings** to bridge this gap.



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Proposed Contributions



Contributions Overview

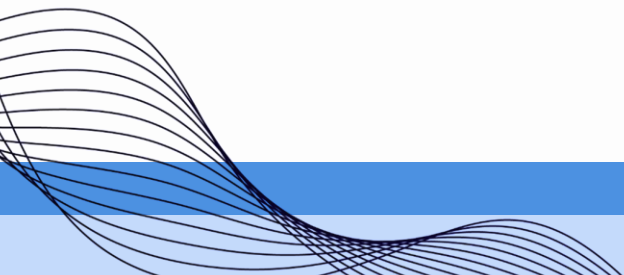
Implemented RotatE-CT and MRotatE

Designed and tested:

- Relation-Balanced Sampling (not adopted)
- Type-Constrained Negative Sampling (not adopted)

Proposed:

- Adversarial Temperature Annealing for stable training
- Combined methods → better link prediction + interpretability



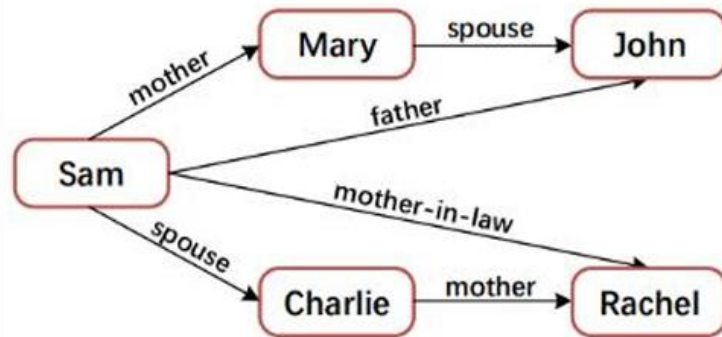
RotatE-CT

translation bias:

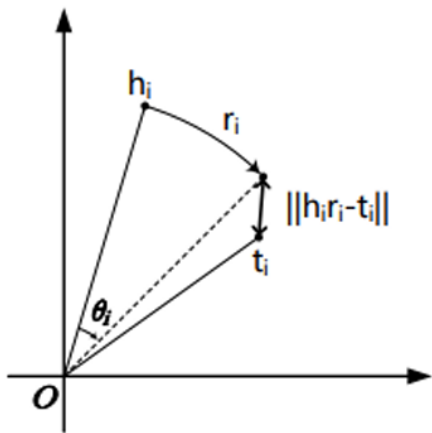
$$t = h \circ r + b_r$$

Breaks commutativity to capture order-sensitive relations.

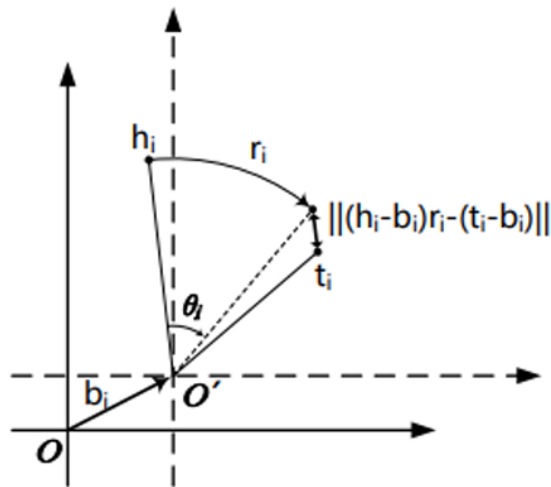
Improves modeling of composition like:
Mother of spouse $\neq$ Spouse of mother



RotatE-CT



(a) RotatE



(b) RotateCT

MRotateE

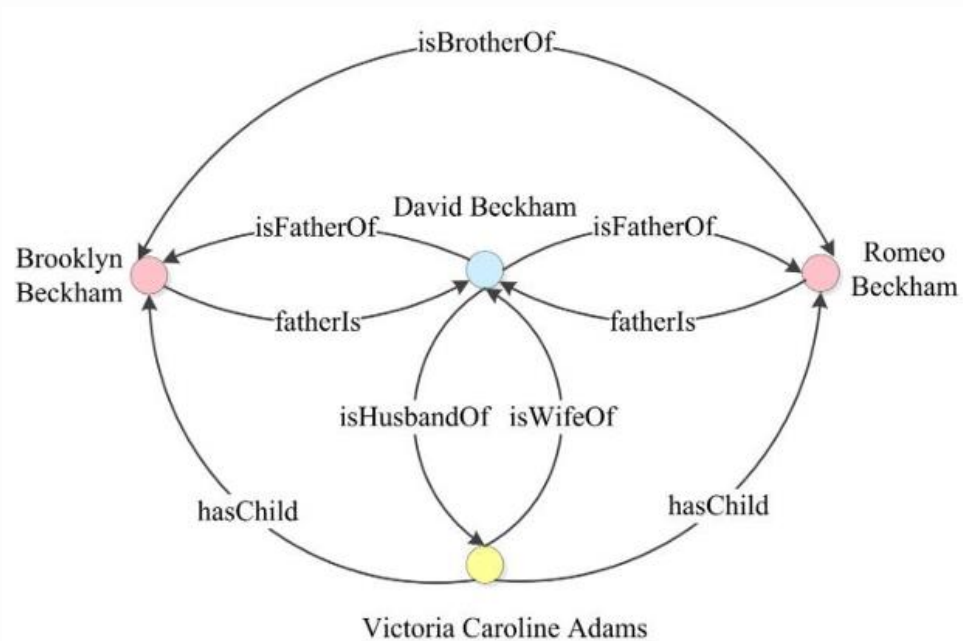
Entity rotation constraint:

$$\|h_e - t_e\| = \|r_e\|$$

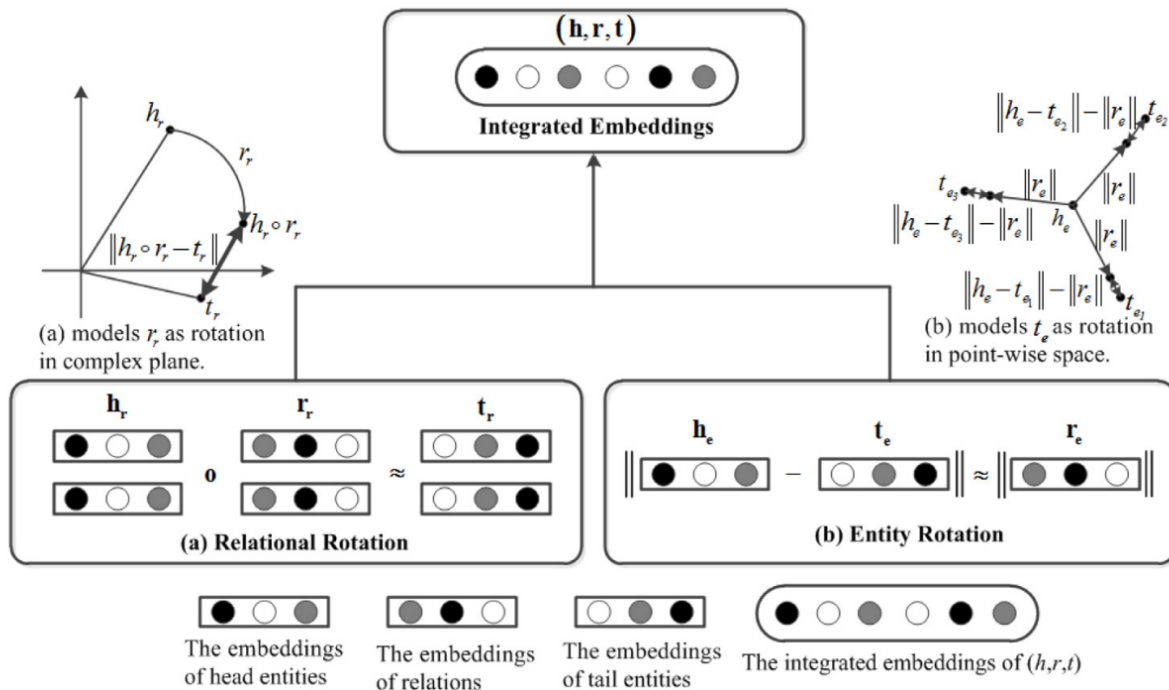
Dual-geometry embeddings (entities rotate around each other).
Handles 1-to-N, N-to-1, N-to-N relations effectively.



MRotateE



Model Framework of MRotateE



Experimental Ideas (Not Adopted)

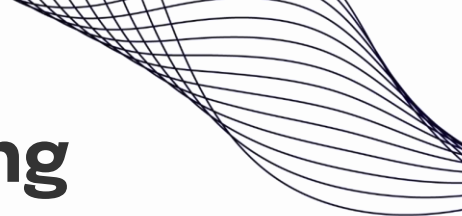
Relation-Balanced Sampling:

- Sample relations uniformly to reduce bias.
- No gain on FB15k dataset.

Type-Constrained Negative Sampling:

- Discard implausible negatives.
- Redundant with adversarial sampling.

Adversarial Temperature Annealing



Adversarial sampling weight:

$$P(i) \propto \exp(\alpha \cdot \text{score}_i)$$

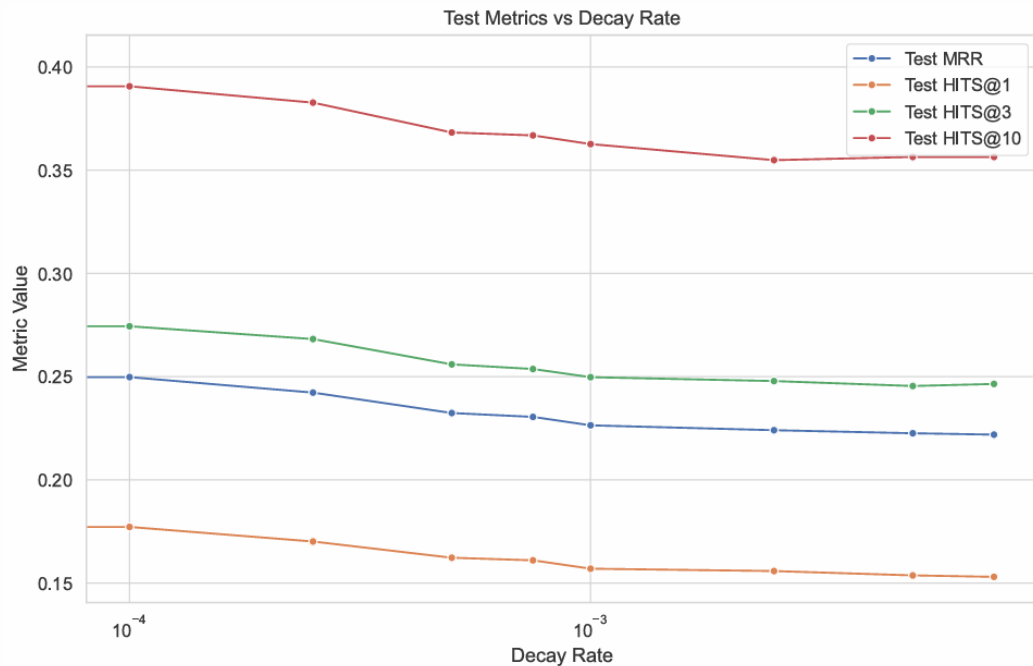
Problem:

Static α = unstable training.

Solution:

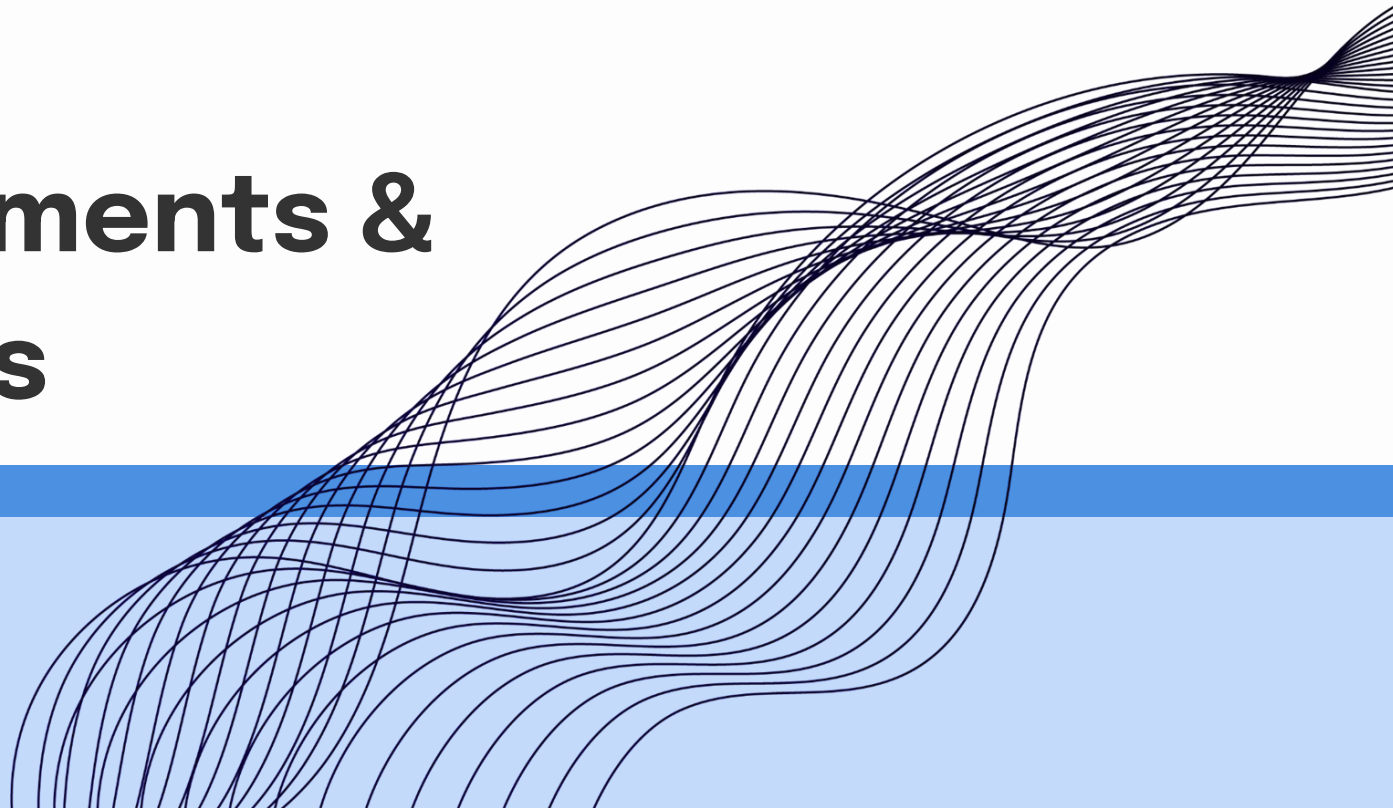
- Diverse negatives $\rightarrow$ Focus on hard negatives.
 - Change temperature over training iterations
- 

Adversarial Temperature Annealing



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Experiments & Results



Ablation Results

- Combined model achieves +40% MRR and -75% MR vs baseline.

ABLATION STUDY RESULTS ON FB15K-237 AT STEP 4999

Model	MRR	MR	HITS@1	HITS@3	HITS@10
RotatE (Base)	0.2175	1010.86	0.1493	0.2395	0.3536
RotatE-CT	0.2748	325.95	0.1899	0.3062	0.4417
MRotatE	0.2964	401.67	0.2100	0.3260	0.4708
RotatE-CT + MRotatE	0.3069	252.93	0.2149	0.3411	0.4892

Summary of Methods

- RotatE-CT Non-commutative relations Rotation + translation bias
- MRotatE Multi-fold relations Entity rotation, dual-geometry
- Adversarial Temp Annealing Static neg. sampling focus Dynamic α schedule

SUMMARY OF PROPOSED ENHANCEMENTS TO ROTATE.

Model / Idea	Limitation Addressed	Mechanism
RotatE-CT	Non-commutative relations	Rotation + translation bias
MRotatE	Multi-fold relations	Entity rotation, dual-geometry embeddings
Relation-Balanced Sampling	Relation frequency imbalance	Balanced relation-level triple sampling
Adversarial Temp Annealing	Static negative sampling focus	Dynamic temperature schedule for negatives

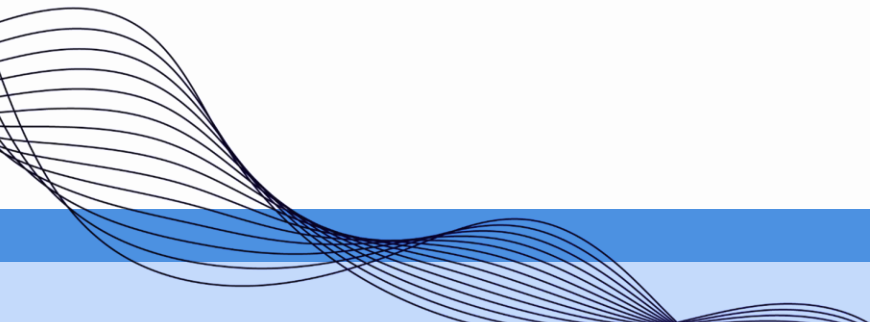
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Conclusion & Future Directions



Conclusions

- Achieved **significant gains** in both accuracy (MRR $\uparrow$ 40%) and efficiency (MR $\downarrow$ 75%) compared to the baseline.
- **Interpretability preserved** through geometric embedding framework, making the models suitable for high-stakes domains (e.g., biomedical KGs).
- Demonstrates that **rotational embeddings are flexible and extensible**, bridging the gap between



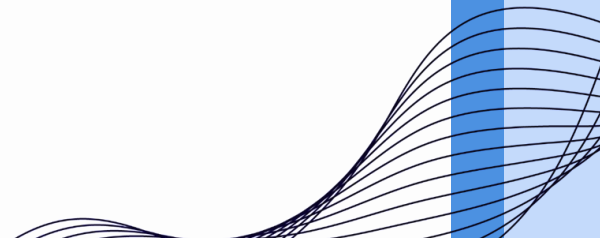
Future Work

Hybrid models → combine geometric embeddings with symbolic reasoning.

Scalability → distributed training & parameter-efficient strategies for large KGs.

Advanced sampling & auto-tuning → improve performance & reduce manual tuning.

Temporal & dynamic KGs → capture evolving relational patterns.



Thanks!

Do you have any questions?

